

Assessing Public Perception of AI-Generated Social Media Content around the 2024 U.S. Presidential Debate

Presenting: Rebecca Ansell,

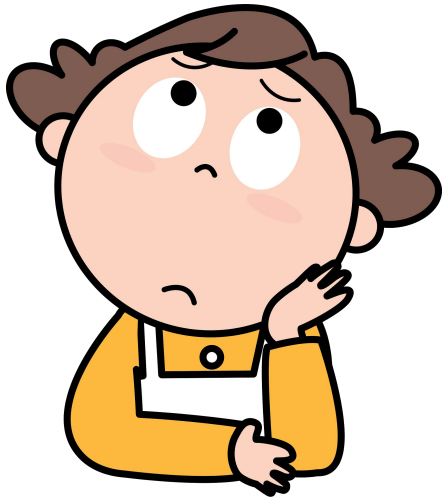
Team Members: Dr. Tiago Ventura, Dr. Sejin Paik, Autumn Toney-Wails, Dr. Leticia Bode, Dr. Lisa Singh

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Research Questions

1. Can humans effectively identify AI-generated content from human-authored content on social media?

2. What characteristics do posts have that are perceived as human?



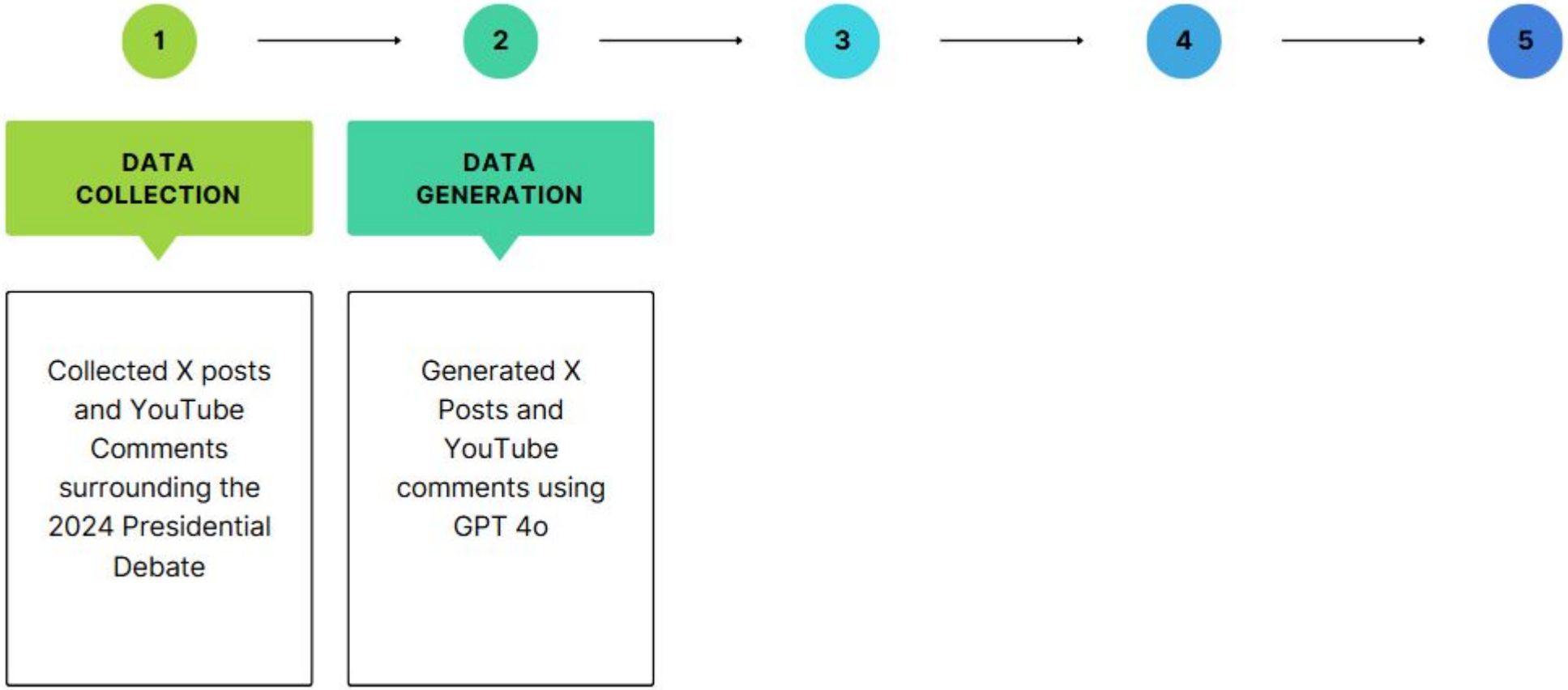
Methodology



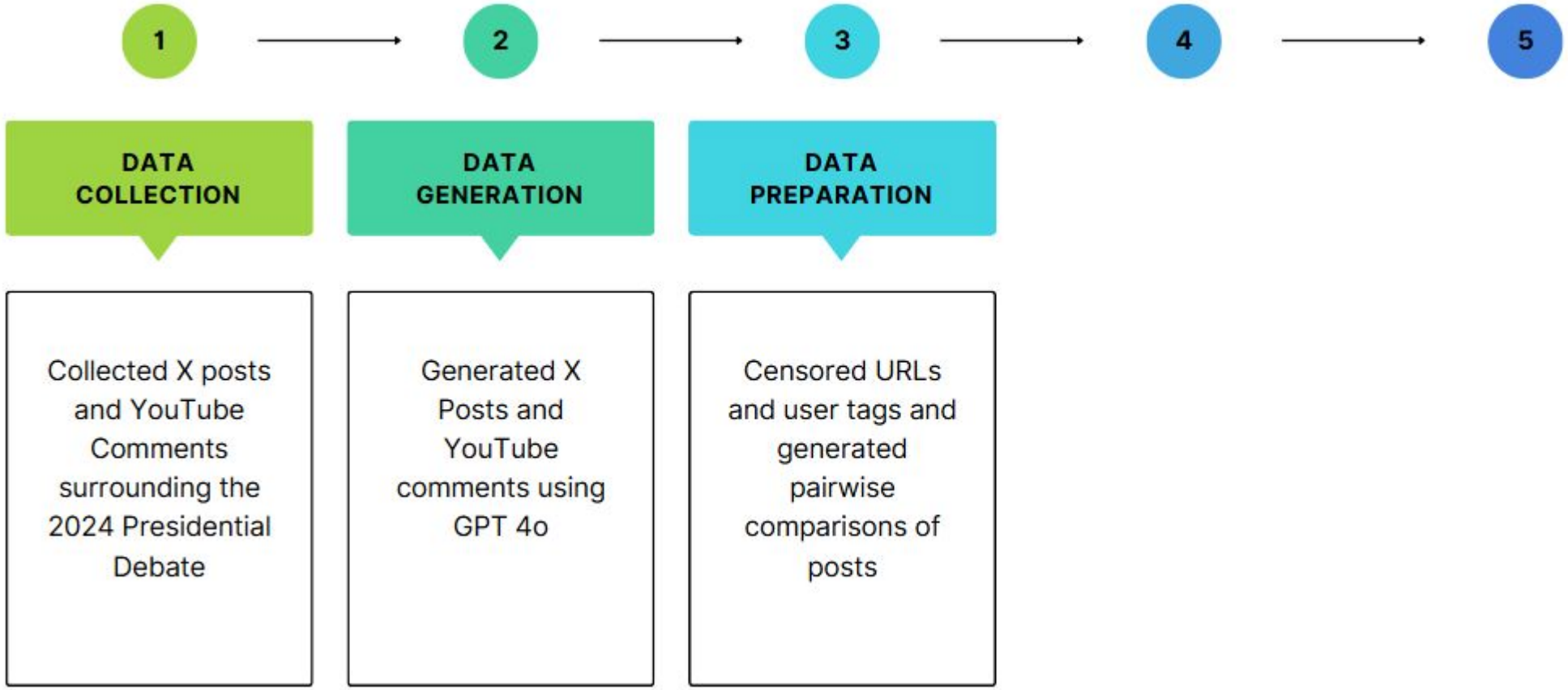
DATA COLLECTION

Collected X posts and YouTube Comments surrounding the 2024 Presidential Debate

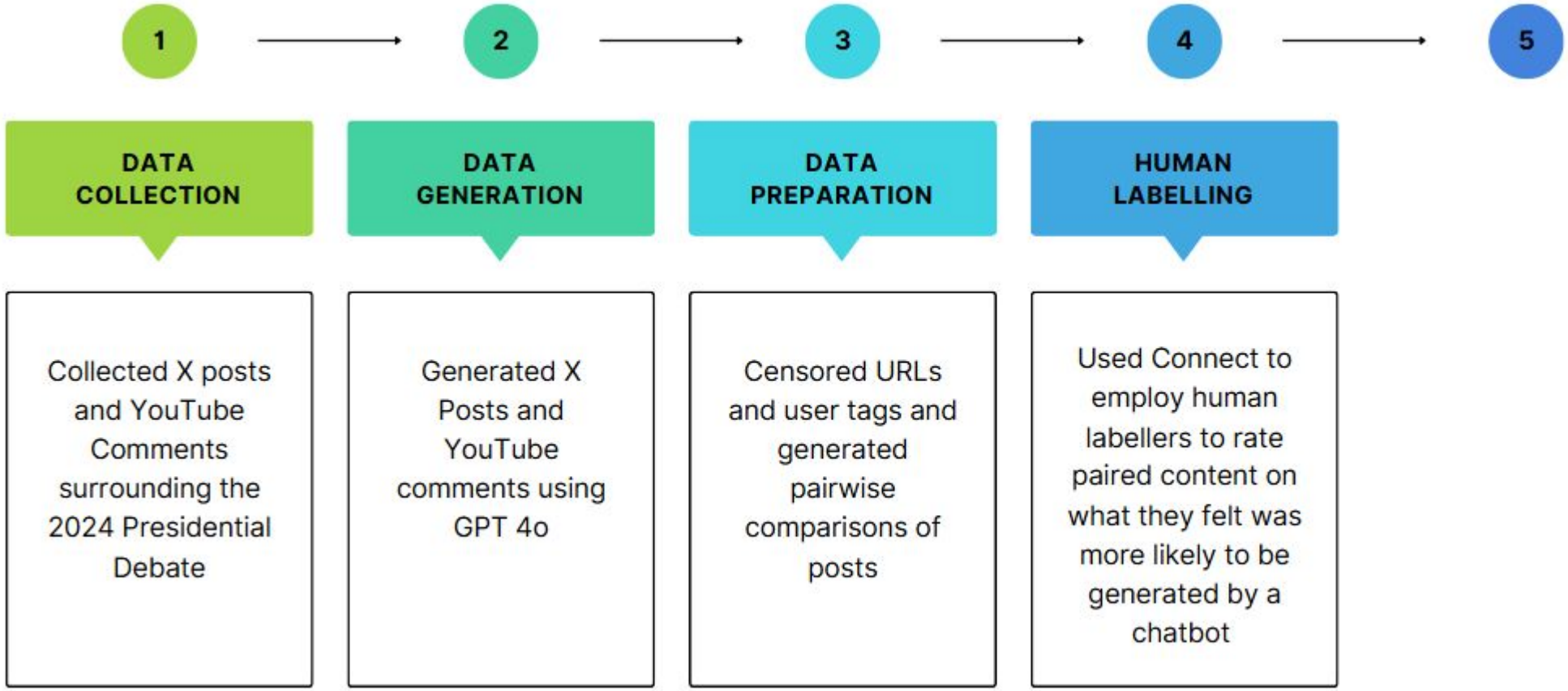
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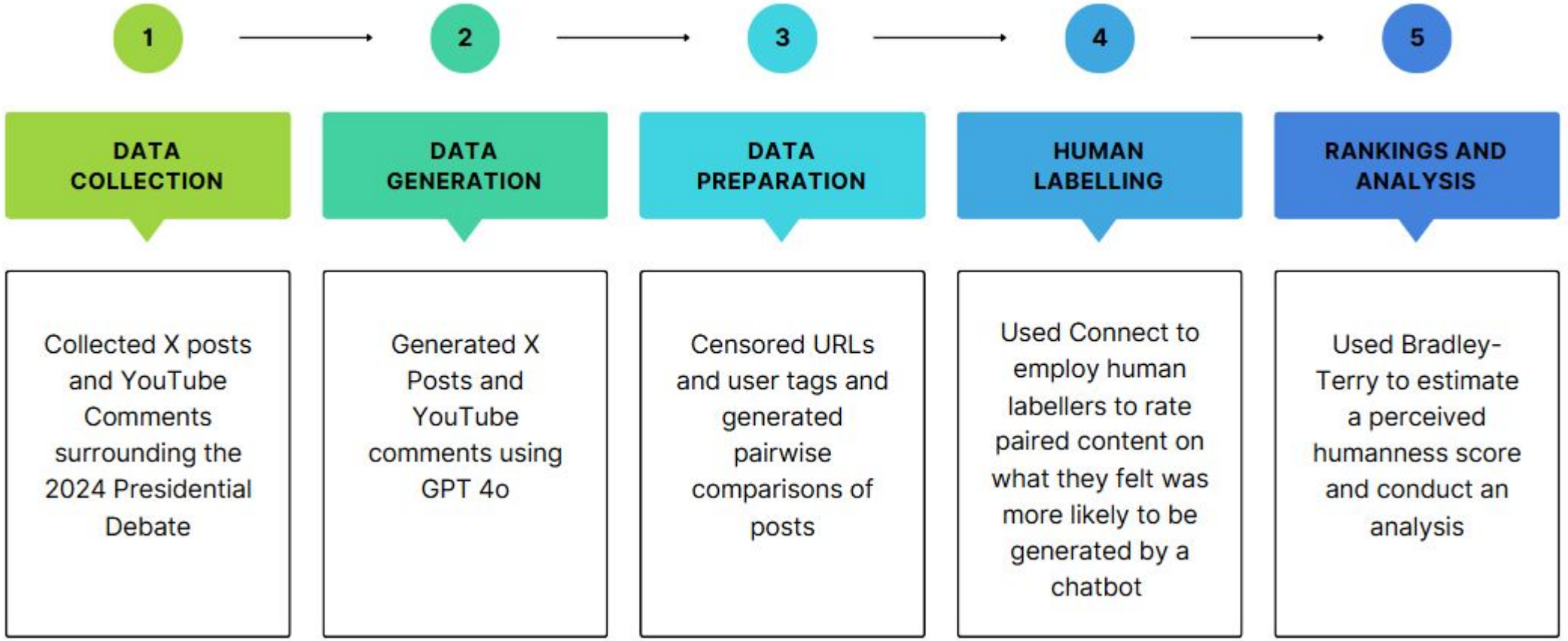
Methodology



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Example Labelling Task 1:

Which of the following 2 X Posts is more likely to be generated by an AI Chatbot?

1. "why do politicians always avoid direct answers?? 😒 #Debatenight"

2. Tonight's #debate will air live on ABC and can be live-streamed on ABC News Live, #Disney+ and #Hulu at 9 p.m. EDT. #DebateNight https://t.co/xzfcgwsr

*these posts are random and in no way reflect my political views or opinion

Example Labelling Task 1:

Which of the following 2 X Posts is more likely to be generated by an AI Chatbot?

1. "why do politicians always avoid direct answers?? 😒 #Debatenight" - X Post

2. Tonight's #debate will air live on ABC and can be live-streamed on ABC News Live, #Disney+ and #Hulu at 9 p.m. EDT. #DebateNight https://t.co/xzfcgwsr - ChatGPT 4o

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Example Labelling Task:

Which of the following 2 YouTube Comments is more likely to be generated by an AI Chatbot?

1. ABC just embarrassed themselves. I had to just stop. They are children as the democrats are.

2. ABC News trying to fact-check Trump was hilarious. The dude was dropping lies left and right! 😂

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Which of the following 2 YouTube Comments is more likely to be generated by an AI Chatbot?

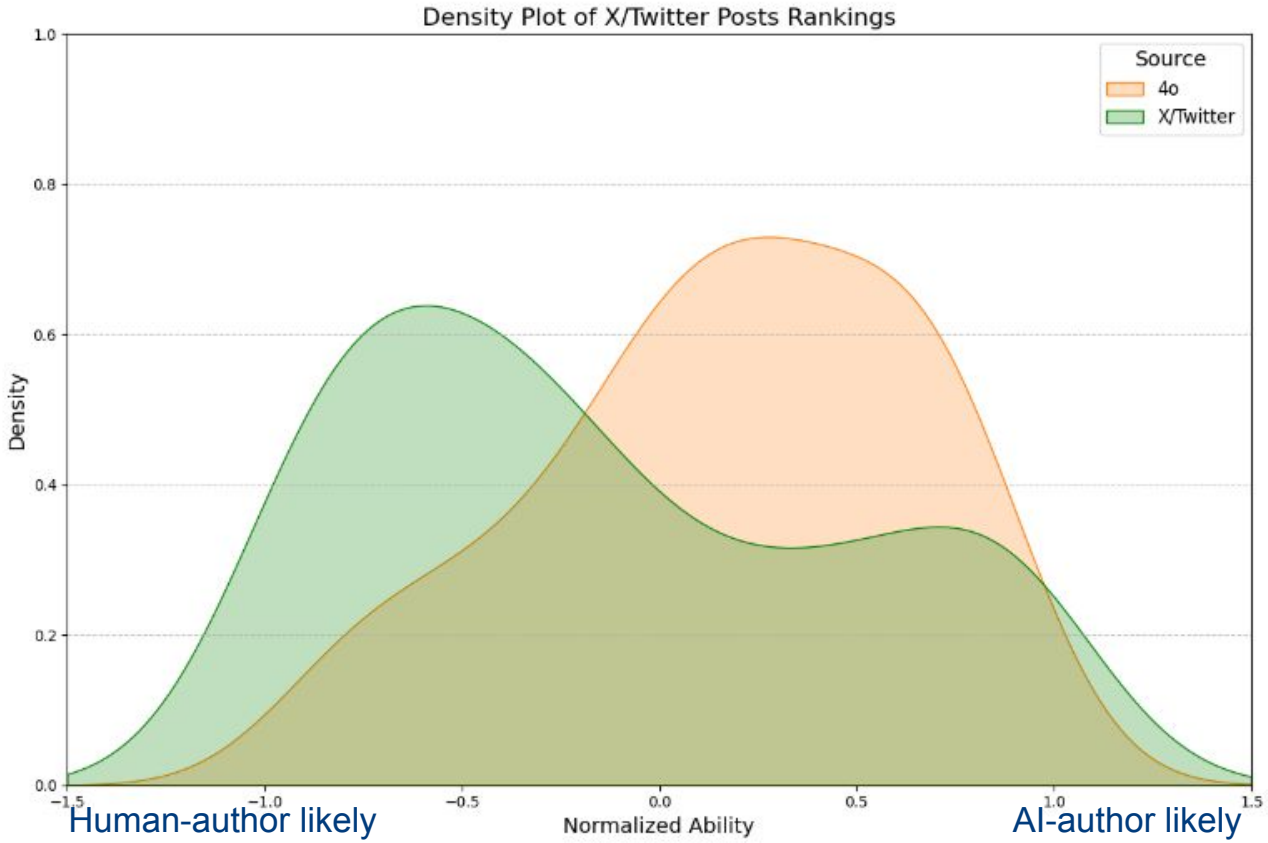
1. ABC just embarrassed themselves. I had to just stop. They are children as the democrats are. - YouTube

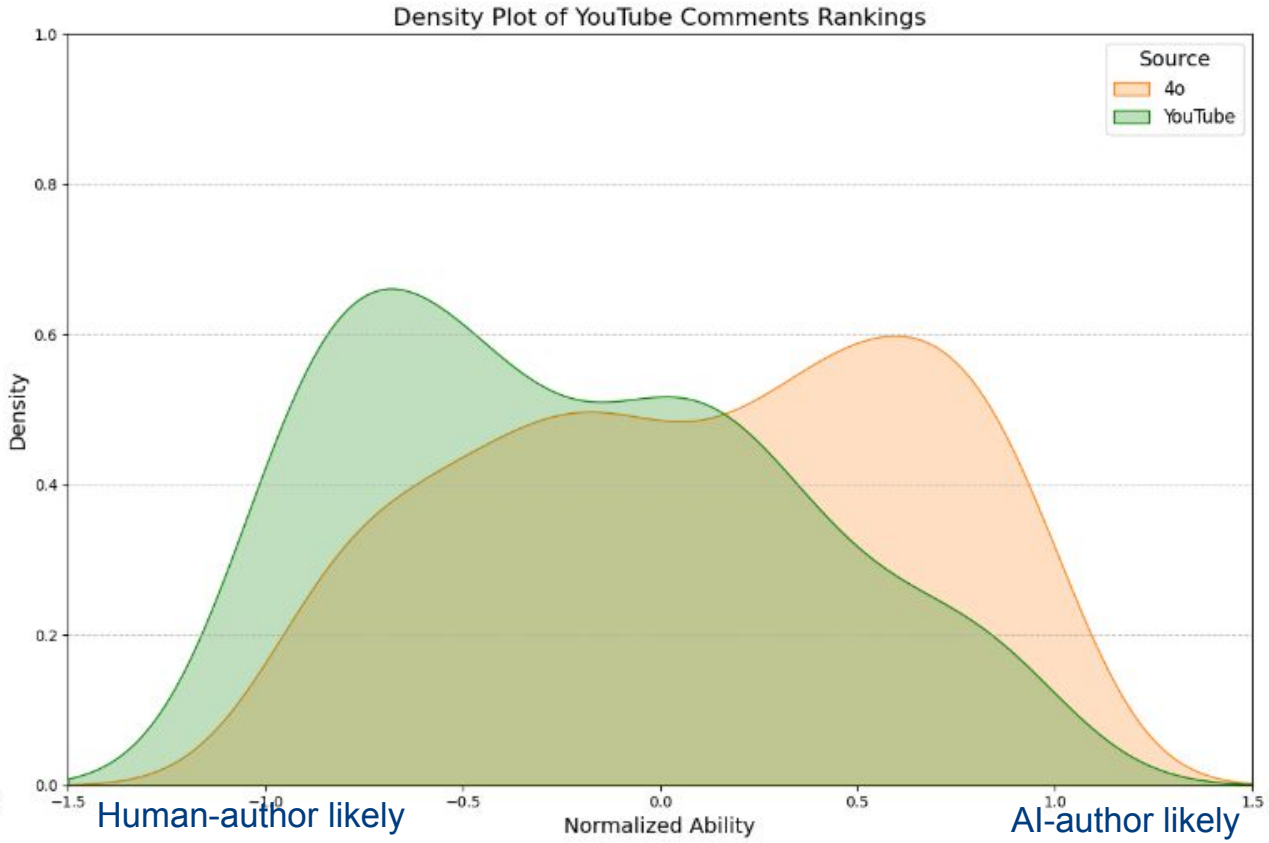
2. ABC News trying to fact-check Trump was hilarious. The dude was dropping lies left and right! 😂 - ChatGPT 4o

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Results

- ➔ Labellers could generally distinguish between AI-generated and human-authored content
- ➔ Separation on X is more than on YouTube

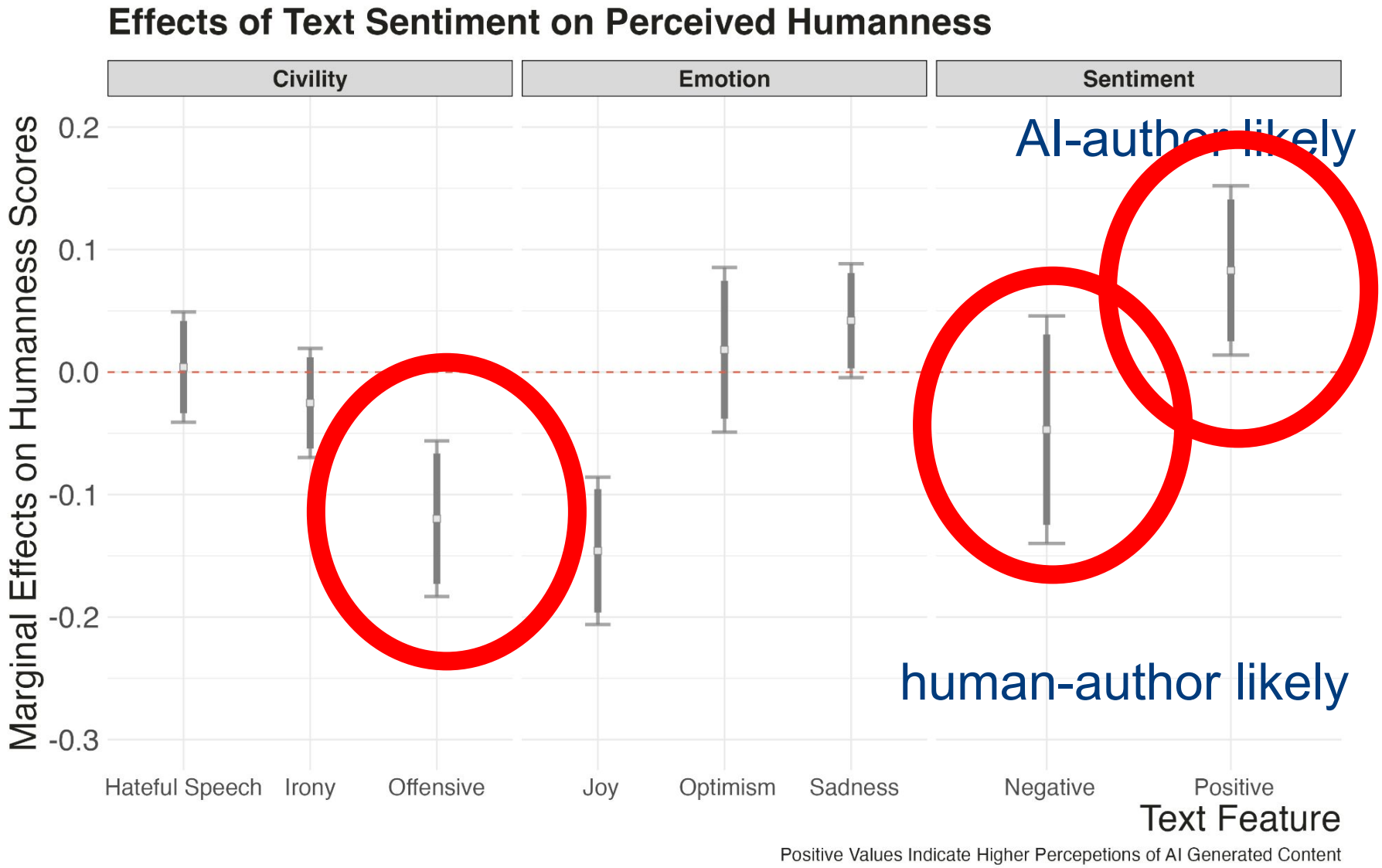




Results

Positive sentiment often perceived as AI-authored.

Negative sentiment and offensive language linked to human authorship.



Conclusion

1. **Humans can** distinguish authorship overall.

2. **Findings illustrate** tone, civility, and emotion markers are factors.

3. **Future work** will include more platforms and LLMs to compare.

Thank you!

Questions: rja80@georgetown.edu



Special thank you to MDI and the TPP Grant for funding this research.

Publication link:

https://www.electionanalysis.ws/us/president2024/section-6-digital-campaign/how-human-analyzing-the-perceived-humanness-of-ai-generated-social-media-content-around-the-u-s-2024-presidential-debate/

Methodology

Data Collection:

- Dataset includes authentic user-generated content relating to the presidential debate and GPT-4o generated posts reflecting diverse political personas.
- GPT-4 utilized debate transcripts as contextual references for generating posts.
- 504 participants annotated 7,500 pairwise comparisons across X (formerly Twitter) and YouTube posts.

Analysis Approach:

- Bradley-Terry model employed to compute perceived humanness scores.
- Scores normalized on a scale from -1 (perceived as human) to 1 (perceived as AI).

Platforms:

- Equal representation of posts from X and YouTube.

